

Saudi Mais Co, for Medical Products
Work Instruction

Production & Test Procedure for Hypodermic Needle

Doc. #	:	WI-8-2-2-04
Rev. #	:	3
Date.#	:	10.12.2022
Page.#	:	1 / 4

1. Purpose:

To stipulate the required tests for sterile Hypodermic Needle to ensure good quality products before releasing to the market.

2. Scope:

Medical Device – **Hypodermic Needle.**

3. Definition:

Hypodermic Needle : A thin hollow needle used with a syringe, suitable for administering hypodermic injections.

4. Responsibilities:

- 4.1 Quality Controller is responsible for sampling & testing.
- 4.2 Quality control supervisor are responsible for checking the obtained results.
- 4.3 Machine operators and assembly peoples are responsible to produced good quantity and good quality products

5. Procedure:

5.1 Receiving Bulk Product:

- Ensure the receiving slip from supplier is correct as per our PO.
- Inspect the carton condition, must be in good condition, no damaged at all cartons with products.
- Quality controller is responsible to check the products as per inspection reports .
- Needle should be with smooth surface, straight, accurate in dimension, with sharp bavel, should be shrp tip and no tip bend, strong gluing between needle and holder, no pull out and record the results in the inspection reports.
- Take samples & inspect in order to AQL procedure, product description, specification with apply the AQL sampling plans for normal inspection.
- **Flow Rate**
When tested in accordance with the standard ISO7864 Annex C, the results should be complying with the requirement of standard.
- **Penetration Force**
When tested in accordance with the standard ISO7864 Annex D, the results should be complying with the requirement of standard.
- **Needle Bonding Strength**
When tested in accordance with the standard ISO7864 Annex E, the results should be complying with the requirement of standard.

Prepared by: Q.A.C
M AZEEM

Approved by : Q.C.M
Mr. Javeed Gafoor

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Page.#	:	2 / 4

- Record into OBM inspection report.

5.2 Visual Inspection.

Visual Appearance:

When inspected by normal or corrected-to-normal vision without magnification under an illuminance of 300 lx to 700 lx, the surface of the hypodermic needle tube should appear free from particles and extraneous matter.

When examined under x2,5 magnification, the hub socket should appear free from particles and extraneous matter.

- The needle surface must be smooth with siliconized.
- Needle should be free from any irregularities, black spot, tip bend, damaged, etc.
- Needle bevel must be smooth & round edge.
- The hub must be good fitting in syringe nozzle, no loose fitting.

Color Coding

The hypodermic needle should be Colour coded in accordance with standard ISO 6009.

Gauge	Color
G18	Pink
G19	Cream
G20	Yellow
G21	Green
G22	Black
G23	Blue
G24	Crimson
G25	Orange
G26	Brown
G27	Grey

Table 1: Hypodermic Needles Dimension:

Gauge	Needle Length ± 0.04" (Inch)	OD ± 0.02 mm	Needle Hub Color	Needle Protector Length ± 1.0 mm	A ± 0.1 mm	B ± 0.1 mm
G18	1" - 1½"	1.25	Pink	42.0 - 54.0 - 62.0	3.0	5.4
G19	1" - 1½" - 1½"	1.10	Cream	42.0 - 54.0 - 62.0	2.95	5.0
G20	1" - 1½" - 1½" - 1½"	0.90	Yellow	42.0 - 54.0 - 62.0	2.85	4.6
G21	1" - 1½" - 1½"	0.80	Green	42.0 - 54.0 - 62.0	2.75	4.2
G22	1" - 1½" - 1½"	0.70	Black	42.0 - 54.0 - 62.0	2.55	3.2
G23	1" - 1½" - 1½"	0.65	Blue	42.0 - 54.0 - 62.0	2.45	3.2
G24	1" - 1½"	0.55	Crimson	42.0 - 54.0 - 62.0	2.45	2.5
G25	¾" - 3" - 1½"	0.50	Orange	42.0 - 54.0 - 62.0	2.45	2.5
G26	¾" - ½"	0.45	Brown	42.0 - 54.0 - 62.0	2.35	2.0
G27	½" - ½"	0.40	Grey	42.0 - 54.0 - 62.0	2.35	2.0

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Page.#	:	3 / 4

6. Packaging and Labeling:

- Production controller is responsible to issue production order for packaging.
- Machine operator is responsible to fix die with approved label and set approved parameters for packaging.
- Each product package in blister pack, the label contains the following.
- Product name and code no., reference no., size, lot no., type, temperature, logo, sterilization logos, production date, expiry date, etc.
- Quality controller will responsible to get the packed sample as per AQL procedure and make sure that package forming, sealing, printing and information are correct.
- Every 2 hours quality controller will responsible to get sample as per AQL sampling plan and record the results in in process control form.
- Then to pack in specified small boxes, and arrange in big carton.
- The big carton must be with label and detail information.

7. Sterilization:

- The finished products are sterilized as per work instruction WI-7-5-1-1.

8. Final Inspection and Release

- The Quality controller must be check, the unloaded products after sterilization .
- Quality controller will responsible to take sample as per AQL smapling plan and inspect the prdoucts as requiremnts mention in final control.
- The quality assurance manager or quality control manager have to review the completed cycle parameters and inspection reports both documents have to be sign.
- After confirmation all the test from the Lab, to release the sterilized products, the document once again have to review the quality control manager then sign.
- The production department responsible to record the quantity of products before transferring to the finished goods ware-house and then it could be released to the customer.

8 Sampling:

- Inspection & Test report for Hypodermic Needle : F-823-T37
- In-process control for packaging machine and final Checking : F-823-i4
- Final inspection and Release Record : F-8-2-4

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Page.#	:	4 / 4

10. Reference:

ISO 7864-1

ISO 6009:1988

11. Modification / Documentation change History:

Date	Changes	Rev
16.10.2018	Point 4.3, 5.2, 5.4, 5.6, & 7 are modified & point 8 added	1
14.06.2020	5.1, 5.2, 9, and 10 updated	2
10.12.2022	Procedure 5.1, 5.2 & 6 updated	3

Prepared by: Q.A.C
MAZEEM

Approved by : Q.C.M
Mr. Javed Gafar